

Project Proposal

Title

Human-in-the-Loop Agentic AI Job Search Copilot

Project Overview

The Human-in-the-Loop Agentic AI Job Search Copilot is an intelligent multi-agent career assistance platform designed to support job seekers throughout the complete job application lifecycle.

The system combines Agentic AI principles, Multi-Agent Architecture, LangGraph orchestration, Retrieval-Augmented Generation (RAG), and Human-in-the-Loop decision making to create an interactive and collaborative career assistant.

Unlike a traditional AI chatbot, the proposed system uses a Supervisor/Orchestrator Agent that coordinates specialized AI agents, validates outputs, manages workflow execution, routes requests, and requests user approval before important decisions are executed.

The project aims to demonstrate practical implementation of advanced Agentic AI concepts while solving a real-world problem faced by job seekers.

Problem Statement

Job seekers often spend significant time searching for suitable job opportunities, comparing their skills against job requirements, customizing cover letters, preparing for interviews, and tracking applications.

Existing job platforms provide job listings but offer limited personalized guidance, skill-gap analysis, and decision support.

This project aims to automate and streamline these activities through a collaborative AI system while keeping the user actively involved in important decisions.

Project Objectives

The system will:

1. Analyze uploaded CVs and professional profiles.
2. Discover relevant job opportunities.
3. Match candidate profiles against job requirements.

4. Identify skill gaps and improvement areas.
 5. Generate customized cover letters.
 6. Prepare interview questions and coaching material.
 7. Track job applications and progress.
 8. Allow users to approve, reject, or modify AI-generated recommendations.
 9. Demonstrate orchestration of multiple specialized AI agents.
 10. Support both cloud-based and offline LLM execution.
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Human-in-the-Loop Design

A key feature of the project is Human-in-the-Loop interaction.

The AI system will not autonomously execute critical decisions. Instead, users will be asked to review and approve recommendations generated by the agents.

Examples include:

- Selecting preferred job opportunities
- Approving skill-gap recommendations
- Reviewing generated cover letters
- Choosing interview preparation modes
- Confirming application status updates

This design ensures transparency, user control, and collaborative decision-making.

System Modes

Mode 1 – Manual Job Matching

The user uploads a CV and manually provides a job description from:

- LinkedIn
- StepStone
- Company career portals
- PDF job advertisements

The system will:

- Analyze the CV
- Parse the job description
- Calculate compatibility score
- Identify missing skills
- Generate recommendations
- Create a tailored cover letter

- Generate interview questions
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Mode 2 – Live Job Discovery

The system will connect to public job APIs to retrieve active job opportunities.

Users can search using:

- Desired job title
- Skills
- Location
- Remote preference

The system will:

- Fetch jobs from public APIs
 - Rank opportunities based on suitability
 - Present recommendations
 - Allow users to select preferred jobs
 - Generate application materials
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Multi-Agent Architecture

The platform will consist of the following specialized agents:

1. Supervisor / Orchestrator Agent

Responsibilities:

- Manage workflow execution
- Route tasks between agents
- Maintain workflow state
- Trigger human approval checkpoints
- Handle exception management

2. Human Interaction Agent

Responsibilities:

- Collect user decisions
- Manage approval requests
- Capture user feedback
- Store preferences

3. CV Analyzer Agent

Responsibilities:

- Parse uploaded CVs
- Extract skills
- Extract experience
- Create structured candidate profiles

4. Job Discovery Agent

Responsibilities:

- Query public job APIs
- Retrieve job opportunities
- Normalize job data

5. Job Parser Agent

Responsibilities:

- Parse job descriptions
- Extract required skills
- Extract qualifications

6. Matching Agent

Responsibilities:

- Compare CVs against jobs
- Calculate compatibility scores
- Generate matching explanations

7. Gap Analysis Agent

Responsibilities:

- Identify missing skills
- Recommend improvement areas
- Suggest learning priorities

8. Cover Letter Agent

Responsibilities:

- Generate personalized cover letters
- Adapt content to specific job roles

9. Interview Coach Agent

Responsibilities:

- Generate interview questions
- Prepare technical and HR interview guidance

10. Application Tracker Agent

Responsibilities:

- Track job applications
- Store application status
- Maintain follow-up history

11. Verification Agent

Responsibilities:

- Validate outputs from other agents
 - Ensure data completeness
 - Detect inconsistencies
 - Improve response reliability
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Workflow Architecture

1. User uploads CV or profile.
 2. Supervisor Agent determines the workflow.
 3. CV Analyzer creates a structured candidate profile.
 4. User selects Manual Mode or Live Search Mode.
 5. Relevant job information is gathered.
 6. Matching Agent calculates compatibility.
 7. Gap Analysis Agent identifies missing skills.
 8. Verification Agent validates outputs.
 9. Human approval checkpoint is triggered.
 10. Cover Letter and Interview Coach Agents generate personalized content.
 11. Application Tracker stores progress.
 12. Results are displayed to the user.
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Technology Stack

Programming Language

- Python

Frontend

- Streamlit

AI Frameworks

- LangGraph
- LangChain

Large Language Models

Cloud Mode:

- OpenAI GPT models

Offline Mode:

- Ollama
- Llama 3.1
- Qwen 2.5

Retrieval-Augmented Generation

- ChromaDB as vector database
- OpenAI embeddings: text-embedding-3-small
- Offline embeddings: nomic-embed-text through Ollama

Storage

MVP:

- JSON-based storage

Future Enhancement:

- SQLite
- PostgreSQL

APIs

- OpenAI API
- Arbeitnow Job API
- Adzuna API optional

Demonstration Dataset

To demonstrate the system across multiple domains, a sample dataset containing at least ten professional profiles and CVs will be created.

Example domains:

- RF Planning Engineer
 - RAN Optimization Engineer
 - Python Developer
 - Data Analyst
 - DevOps Engineer
 - Project Manager
 - HR Specialist
 - Marketing Specialist
 - Customer Support Specialist
 - Finance Specialist
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Expected Learning Outcomes

This project demonstrates practical application of:

- Agentic AI Systems
 - Multi-Agent Collaboration
 - Workflow Orchestration
 - Human-in-the-Loop AI
 - Retrieval-Augmented Generation
 - Tool Calling
 - API Integration
 - Local and Cloud LLM Deployment
 - State Management
 - AI System Design
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Expected Deliverables

1. Fully functional Streamlit application.
 2. LangGraph-based multi-agent workflow.
 3. Human-in-the-Loop approval system.
 4. Manual job matching functionality.
 5. Live job discovery functionality.
 6. RAG implementation using ChromaDB.
 7. Cover letter generation module.
 8. Interview preparation module.
 9. Application tracking module.
 10. OpenAI and Ollama support.
 11. Technical documentation and architecture diagrams.
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Expected Outcome

The final system will function as a collaborative AI-powered career assistant capable of coordinating multiple specialized agents while maintaining human oversight and control.

The project will showcase advanced Agentic AI design patterns and demonstrate a practical real-world application of Multi-Agent AI systems for job search, career planning, and application preparation.